CYLINDER LINERS

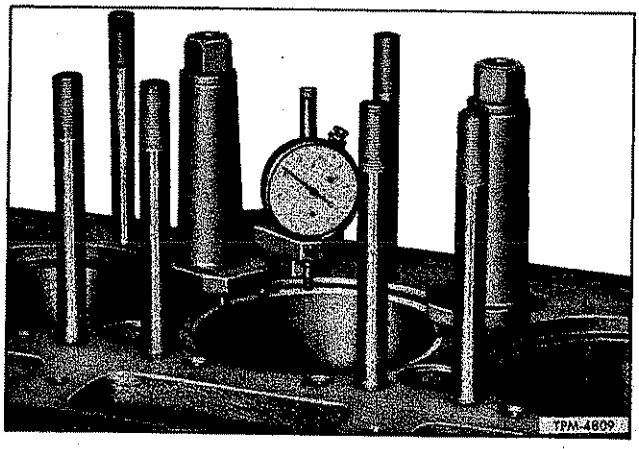


Figure 13—Checking Depth of Liner Below Top of Block

FINISH HONING AND FITTING LINER

Rough hone cylinder bore until the liner can be pushed from 3″ to 4″ into the bore, or until bore is within approximately .0005″ of the diameter of the liner. Do not expect finishing stones to remove more than .001″ of stock, or to true up the bore to any extent.

Work the finish hone with short, rapid strokes up and down the bore. Use light tension on the stones and hone only enough to allow the liner to enter the bore either with a light push fit or a free slip fit.

If new liners are to be used after honing the block, follow instructions for replacing loose liners. If old liners requiring rehoning on the inside diameter are to be used, follow instructions under "Honing Liner" previously in this section.

LINER INSTALLATION

Check bore in cylinder block. Bore must be round and smooth within .0015″ from top to bottom. When checking for these conditions, use a dial indicator, or if a new, undamaged (round) liner will slip fit into the bore, the fit is satisfactory. Hone bore as outlined under "Honing Cylinder Block" previously in this section to remove roughness or distortion.

Before installing the liner or insert in cylinder block wipe the liner, insert and block counterbore until absolutely clean. Use a fine flat honing stone to remove any burrs that may be present on the liner or insert or in cylinder block counterbore. Liner must be installed dry—do not use oil.

Since cylinder head and cylinder block are metal-to-metal contact, each cylinder liner is sealed individually by compression seals. This construction requires that top of liner flange be .0465″ to .050″ below top of cylinder block.

When insert and cylinder liner are installed, height of liner must be checked to determine that liner is proper distance below top of cylinder block. Install hold-down tool over top of liner to hold liner tightly in place, while measuring liner depth below top of block (fig. 13). Use a dial indicator and a surface plate on a flat surface gauge to measure depth of cylinder liner below top of cylinder block.

LINER HEIGHT

When cylinder liner and insert are installed and liner depth is checked in manner previously outlined the depth should be .0465″ to .050″ as indicated. If liner depth is not within these limits, and inspection indicates they are properly installed, the insert or liners or both should be removed for careful inspection of mounting surfaces and condition of cylinder block counterbore. Undersize cylinder liner inserts are available for use when top of cylinder block has been "trued up" by removing a small amount of stock. **DO NOT USE SHIMS TO OBTAIN CORRECT LINER HEIGHT.**

Having replaced the cylinder liners, the pistons may be fitted and the engine rebuilt. In this process, see "Fitting Pistons," in PISTON AND CONNECTING ROD section later in this manual, also refer to the various other sections of this text for detailed instructions relative to correct location of parts, bearing clearances, etc.

Operate engine as instructed in ENGINE RUN-IN PROCEDURE section later in this manual.